CALCULUS · UNIT 1: PRECALCULUS REVIEW

Phone Battery Lab

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students test a smartphone. Battery drain compares linear and exponential models, charging uses an exponential approach to 100%, the data plan is piecewise, conversions use composition and inverses, and screen brightness is sinusoidal.

Average and instantaneous rates of change preview the derivative.

Standards Addressed

- AP CED Unit 0: function families and transformations
- F-BF.B.4 Inverse functions; F-LE.A.4 logarithms
- F-TF.B.5 Sinusoidal models
- Average rate of change (preview of 2.1)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: "How long until your phone dies?" Part 1 models.
Day 2	Part 2 piecewise, compositions, and inverses.
Day 3	Part 3 brightness and rates; lab report.

Materials

- Student handout and Lab Data Cards
- Graphing calculators or Desmos
- Optional: students' own battery data

How the Three Versions Differ

Level	What changes
CP2	Linear vs. exponential drain, piecewise plan, conversion inverse, and log solving.
CP1	Intersections, asymptotes, compositions, transformations, rates of change, and a cosine model.
Honors	Proofs about asymptotes, difference quotients, composite models, continuous rates, and sinusoidal rates.

Watch For (Common Misconceptions)

- Using 0.18 instead of 0.82 for an 18% decrease.
- Forgetting that ln undoes e only when e is isolated.
- Calculator in degree mode for trig models.

Extension Ideas

- Record your own phone's battery for a day.
- Model a laptop battery.
- Compare fast and standard chargers.

Answer Key – CP2

Part 1: Battery Models

#	Task	Answer
1	Write a linear model for streaming: S(t) = 95 − 12t. When is the battery at 20%?	95 − 12t = 20 → t = 6.25 h
2	Write an exponential model for gaming starting at 95%.	G(t) = 95(0.82)^t
3	Find G(3) and S(3). Which drains faster in the first 3 hours?	G(3) ≈ 52.4%, S(3) = 59%; gaming drains faster
4	What is the domain of S in this context?	0 ≤ t ≤ $\frac{95}{12}$ ≈ 7.9 hours

Part 2: Plans & Conversions

#	Task	Answer
5	Write the cost as a piecewise function P(g).	P(g) = 30 for 0 ≤ g ≤ 15; P(g) = 30 + 4(g − 15) for g > 15
6	Find P(12) and P(20).	$30 and $50
7	The phone's temperature is 35°C. Use F = 1.8C + 32 to convert. Then find the inverse function C(F).	95°F; C = $\frac{F-32}{1.8}$
8	Is P(g) one-to-one? Explain.	No; every g from 0 to 15 gives $30.

Part 3: Charging & Brightness

#	Task	Answer
9	Find B(0), B(30), and B(60).	20%, ≈ 76.0%, ≈ 92.7%
10	Solve B(t) = 90 using logarithms.	80e^(−0.04t) = 10 → t = $\frac{\ln 8}{0.04}$ ≈ 52 min
11	Brightness has a maximum 900 at 1 p.m. and a minimum 100 at 1 a.m. Find the midline, amplitude, and period.	Midline 500, amplitude 400, period 24 h

Reflection

#	Prompt	Look for
R1	How can you tell from a table whether data is linear or exponential?	Linear: constant differences. Exponential: constant ratios.
R2	Why does the charging model level off at 100?	e^(−0.04t) approaches 0, so B(t) approaches 100.

Answer Key – CP1

Part 1: Drain & Charge

#	Task	Answer
1	When are the streaming (95 − 12t) and gaming ($95(0.82)^t$) batteries equal after t = 0? Use a graph or table.	They are equal at t = 0 and again near t ≈ 5.0 h (≈ 35.5%); gaming is lower in between, and streaming is lower after that.
2	Solve B(t) = 95 for the charging model.	80e^(−0.04t) = 5 → t = $\frac{\ln 16}{0.04}$ ≈ 69.3 min
3	Describe the end behavior of B(t) and write its horizontal asymptote.	As t → ∞, B → 100; y = 100
4	Write the inverse of B(t) and explain what it tells you.	t = −25 ln($\frac{100 - B}{80}$); the time to reach a given charge

Part 2: Function Toolkit

#	Task	Answer
5	Write P(g) as a piecewise function and graph it. Is it continuous at g = 15?	Yes; both pieces give $30 at g = 15.
6	Let C(F) = $\frac{F - 32}{1.8}$ and W(C) = $0.02C^2$ (heat warning level). Find W(C(104)).	C(104) = 40; W = 0.02(1,600) = 32
7	Describe the transformation from $y = e^t$ to y = 100 − 80e^(−0.04t).	Horizontal stretch by 25 and reflection over the y-axis, vertical stretch by 80, reflection over the x-axis, shift up 100
8	Find the average rate of change of B on [0, 30] and on [30, 60]. What do you notice?	≈ 1.87%/min and ≈ 0.56%/min; charging slows down

Part 3: Brightness Cycle

#	Task	Answer
9	Write a cosine model L(t) for brightness, t in hours after midnight.	L(t) = 400 cos($\frac{\pi}{12}$(t − 13)) + 500
10	Find L(8) and L(20).	L(8) ≈ 400 cos(−$\frac{5\pi}{12}$) + 500 ≈ 603.5; L(20) ≈ 400 cos($\frac{7\pi}{12}$) + 500 ≈ 396.5
11	When is brightness at least 700 nits?	cos(…) ≥ 0.5 → |t − 13| ≤ 4 → 9 a.m. to 5 p.m.

Reflection

#	Prompt	Look for
R1	What does the average rate of change tell you that a single value does not?	How quickly the quantity changes over an interval, a preview of the derivative.
R2	Why is continuity of the data plan important to customers?	No sudden jump in price at 15 GB.

Answer Key – Honors

Part 1: Model Analysis

#	Task	Answer
1	Show algebraically that B(t) never reaches 100.	B = 100 requires e^(−0.04t) = 0, which has no solution.
2	Find the time for B to go from 50% to 90% and compare with 20% to 50%.	50%: t = 25 ln($\frac{80}{50}$) ≈ 11.8 min; 90%: ≈ 52.0 → 40.2 min vs 11.8 min
3	Estimate the instantaneous rate of change of B at t = 30 using a symmetric difference with h = 0.1.	$\frac{B(30.1)-B(29.9)}{0.2}$ ≈ 3.2e^(−1.2) ≈ 0.96 %/min
4	As h → 0, what value do the difference quotients for B at t = 0 approach? Use h = 1, 0.1, 0.01.	3.2 %/min (≈ 3.14, 3.19, 3.20)

Part 2: Composite Models

#	Task	Answer
5	The phone's temperature in °C while charging is C(B) = 25 + 0.12B. Find C(B(t)) and its value at t = 60.	C = 25 + 0.12(100 − 80e^(−0.04t)); C(B(60)) ≈ 36.1°C
6	Find the inverse of S(t) = 95 − 12t and of G(t) = 95(0.82)^t.	$S^{-1}(s) = \frac{95-s}{12}$; $G^{-1}(g) = \frac{\ln(g \div 95)}{\ln 0.82}$
7	Gaming drains 18% per hour. Find the equivalent continuous rate k in G = 95e^{kt}.	k = ln 0.82 ≈ −0.198
8	Explain why the streaming model eventually gives impossible values but the gaming model does not.	Linear decrease becomes negative after 7.9 h; exponential decay approaches 0.

Part 3: Brightness & Rates

#	Task	Answer
9	Find the average rate of change of L from 7 a.m. to 9 a.m. and from 12 p.m. to 2 p.m. Explain the difference.	7–9: (700 − 500) ÷ 2 = 100 nits/h; 12–2: 0 (symmetric about the maximum)
10	Estimate when brightness changes fastest. Justify with the graph.	At 7 a.m. and 7 p.m. (midline crossings), where the cosine is steepest
11	Write the lab report with a model for each phone feature and one question calculus could answer next.	Answers vary.

Reflection

#	Prompt	Look for
R1	How do difference quotients preview the derivative?	As h shrinks, the average rate approaches the instantaneous rate.

#	Prompt	Look for
R2	Why choose exponential rather than linear models for charging?	The charging rate is proportional to how much is left, which produces exponential behavior.